

MODEL SELECTION FOR FUTURE DATA IN THE CASE OF MULTIVARIATE REGRESSION ANALYSIS

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Key words: Adequate subsets, Allen's PRESS, Mallows' C_p^* statistic, model selection, multivariate linear regression, predictive sample reuse, simultaneous test procedures.

Summary: The problem of variable selection in multivariate regression analysis is considered. The method of predictive sample reuse derived by Geisser and Eddy is extended to the multivariate linear regression case. By applying this method to real data it is compared with McKay's "adequate subsets" procedure, Mallows' C_p^* statistic and Allen's PRESS.

1. INTRODUCTION

The techniques of regression analysis have been used widely in the problems of prediction in various disciplines. When the number of variables is large, it is of interest to select a

*Financially supported by the CSIR and the University of the O.F.S. central research fund.

small number of important variables which are adequate for prediction. Whereas the computational aspects of determining a "best" subset in multiple regression have been extensively treated in the literature (see for example Draper and Smith (1981) for a description of the various selection procedures and Hocking (1976) for a useful summary), the topic has not been widely treated in multivariate analysis. Other than the papers by McCabe (1975), Hawkins (1976), McHenry (1978), McKay (1977 and 1979) and Hintze (1980) it seems that little has been done toward determining a "best" subset in multivariate regression analysis.

The first three authors use Wilks' (1932) statistic as the selection criterion while Hintze uses a technique known as "elemental analysis" and shows that the use of elements aids in the selection of variables in multivariate regression or in multivariate discriminant analysis.

McKay on the other hand proposes a number of procedures which are generalizations of Aitkin's (1974) procedure and which are in fact applications of so-called simultaneous test procedures given by Gabriel (1968). Each procedure sets out to find, for each subset of dependent variables, all subsets of independent variables which are as highly correlated with the subset of dependent variables as the original set of independent variables. In fact, selection is based on the squared canonical correlations between independent and dependent variables. A graphical aid to the assessment of the relative importance of subsets of independent variables is also given by McKay (1979).

Geisser and Eddy (1979) also consider the problem of model selection in the case of multiple linear regression. They use the method of predictive sample reuse and a Bayesian predictive density derived from a vague prior distribution to determine which of the different models yields the best predictions for future observations. The model giving the largest likelihood

is selected as the most appropriate of the models being considered.

The main purpose of this note is to extend the work of Geisser and Eddy to the multivariate linear regression case and by using real data, to compare it with McKay's method, Mallows' C_p statistic (1973) and Allen's PRESS (1971).

2. A PREDICTIVE APPROACH TO MODEL SELECTION

The selection of the model that best predicts a future observation using predictive sample reuse has been considered by Geisser and Eddy (1979) for the case of a single dependent variable regressed on several independent variables. Here we shall consider an application of an extension of their method to several dependent variables regressed on several independent variables.

The model in general is:

$$(2.1) \quad E(\underline{Y}|\underline{Z})(q \times 1) = \underline{B}\underline{W}; \quad \underline{W} = \begin{pmatrix} 1 \\ \underline{Z} \end{pmatrix}.$$

$\underline{B} = (\underline{\alpha}, \underline{\beta}) = \begin{pmatrix} \alpha_1 & \beta_{11} \cdots \beta_{1p} \\ \dots & \dots \\ \alpha_q & \beta_{q1} \cdots \beta_{qp} \end{pmatrix}$ is the regression matrix on $\underline{Z}(p \times 1)$.

Let $\underline{Y}^{(N)} = (\underline{Y}_1, \dots, \underline{Y}_N)$ and $\underline{W}^{(N)} = (\underline{W}_1, \dots, \underline{W}_N)$ denote the observations on $\underline{Y}$ and $\underline{Z}$. From Geisser (1965) the predictive density function of a future $\underline{Y}$ given $\underline{W} = \underline{w}$ is given by

$$(2.2) \quad P_{\underline{Y}}(\underline{y}|\underline{w}, \underline{Y}^{(N)}, \underline{W}^{(N)}) = K\{1 + C(\underline{y} - \hat{\underline{B}}\underline{w})' S^{-1}(\underline{y} - \hat{\underline{B}}\underline{w})\}^{-\frac{1}{2}(N-p-1)}$$

where

$$K = (C/\pi)^{\frac{1}{2}q} |S|^{-\frac{1}{2}} \Gamma(\frac{1}{2}(N-p-1)) / \Gamma(\frac{1}{2}(N-p-1-q))$$

$$C = 1 - \underline{w}'(UU')^{-1}\underline{w}$$

$$U = (W^{(N)}, \underline{w})$$

$$\hat{B} = Y^{(N)} W^{(N)'} (W^{(N)} W^{(N)'})^{-1}$$

$$S = (Y^{(N)} - \hat{B}\underline{w})(Y^{(N)} - \hat{B}\underline{w})'$$

The predicted value of $\underline{Y}$ given $\underline{w}$ is given by

$$(2.3) \quad \text{Pred}(\underline{Y}|\underline{w}) = \hat{B}\underline{w}$$

To select the model that best predicts such a future observation is done by using the predictive sample reuse method (Geisser and Eddy (1979)).

Let $Y_{(j)}^{(N-1)} = (\underline{Y}_1, \dots, \underline{Y}_{j-1}, \underline{Y}_{j+1}, \dots, \underline{Y}_N)$ and

$W_{(j)}^{(N-1)} = (\underline{W}_1, \dots, \underline{W}_{j-1}, \underline{W}_{j+1}, \dots, \underline{W}_N)$ denote the

observations on $\underline{Y}$ and $\underline{W}$ (or $\underline{Z}$) with the j -th observations left out. Then the predictive density function of $\underline{Y}_j$ given $\underline{W}_j$ is given by

$$(2.4) \quad P_{\underline{Y}_j}(\underline{Y}_j|\underline{W}_j, Y_{(j)}^{(N-1)}, W_{(j)}^{(N-1)}) \propto \{1 + C_j(\underline{Y}_j - \hat{B}_j \underline{W}_j)' S_j^{-1} (\underline{Y}_j - \hat{B}_j \underline{W}_j)\}^{-\frac{1}{2}(N-p-2)}$$

where

C_j , $\hat{B}_j$ and S_j are defined as in (2.2) but with $Y^{(N)}$ replaced by $Y_{(j)}^{(N-1)}$, $W^{(N)}$ by $W_{(j)}^{(N-1)}$ and $\underline{w}$ by $\underline{W}_j$.

Let $\Omega_{k:m}$ denote the model

$$(2.5) \quad E(\underline{Y}^{(k)} | \underline{W}^{(m)}) = B^{(kxm)} \underline{W}^{(m)}$$

where k and m are subsets of $\underline{Y}$ and $\underline{Z}$ respectively and indicate the number of dependent and independent variables used in the model. $B^{(kxm)}$ is the appropriate matrix of regression coefficients.

Note that equation (2.5) is not a marginal prediction density (i.e., it is not obtained by integrating over the joint predictive density) but it is only (2.4) for various m and k . The reason for this is that in doing model selection on subsets of $\underline{Y}$ and $\underline{Z}$ it is assumed that only the sample values for the subsets are available.

The procedure is then to select the model $\Omega_{k:m}$ that maximizes the likelihood

$$(2.6) \quad L(\Omega_{k:m}) = \prod_{j=1}^N p_{Y_j}(\underline{Y}_j | \underline{W}_j, Y_j^{(N-1)}, W_j^{(N-1)})$$

where the predictive density is calculated according to (2.4) under model $\Omega_{k:m}$.

3. AN APPLICATION OF PREDICTIVE SAMPLE REUSE TO REAL DATA

Consider the following real multivariate regression problem obtained from Anderson and Bancroft (1952) page 205. Twenty-five samples of tobacco leaf were analysed for organic and inorganic chemical constituents, and regression was used to discover the nature and extent of the relationship of certain of these constituents. The dependent variables considered are:

$Y_1 \equiv$ Rate of cigarette burn in inches per 1000 seconds

$Y_2 \equiv$ Per cent sugar in the leaf

$Y_3 \equiv$ Per cent nicotine.

The independent variables are:

$Z_1 \equiv$ Percentage of nitrogen $Z_4 \equiv$ Percentage of phosphorus

$Z_2 \equiv$ Percentage of chlorine $Z_5 \equiv$ Percentage of calcium

$Z_3 \equiv$ Percentage of potassium $Z_6 \equiv$ Percentage of magnesium.

In other words, $p = 6$ and $q = 3$. The original data (see also Groenewald *et al.*, (1980)) are given in the following table:

TABLE 3.1
TOBACCO DATA

Y_1	Y_2	Y_3	Z_1	Z_2	Z_3	Z_4	Z_5	Z_6
1,55	20,05	1,38	2,02	2,90	2,17	,51	3,47	,91
1,63	12,58	2,64	2,62	2,78	1,72	,50	4,57	1,25
1,66	18,56	1,56	2,08	2,68	2,40	,43	3,52	,82
1,52	18,56	2,22	2,20	3,17	2,06	,52	3,69	,97
1,70	14,02	2,85	2,38	2,52	2,18	,42	4,01	1,12
1,68	15,64	1,24	2,03	2,56	2,57	,44	2,79	,82
1,78	14,52	2,86	2,87	2,67	2,64	,50	3,92	1,06
1,57	18,52	2,18	1,88	2,58	2,22	,49	3,58	1,01
1,60	17,84	1,65	1,93	2,26	2,15	,56	3,57	,92
1,52	13,38	3,28	2,57	1,74	1,64	,51	4,38	1,22
1,68	17,55	1,56	1,95	2,15	2,48	,48	3,28	,81
1,74	17,97	2,00	2,03	2,00	2,38	,50	3,31	,98
1,93	14,66	2,88	2,50	2,07	2,32	,48	3,72	1,04
1,77	17,31	1,36	1,72	2,24	2,25	,52	3,10	,78
1,94	14,32	2,66	2,53	1,74	2,64	,50	3,48	,93
1,83	15,05	2,43	1,90	1,46	1,97	,46	3,48	,90
2,09	15,47	2,42	2,18	,74	2,46	,48	3,16	,86
1,72	16,85	2,16	2,16	2,84	2,36	,49	3,68	,95
1,49	17,42	2,12	2,14	3,30	2,04	,48	3,28	1,06
1,52	18,55	1,87	1,98	2,90	2,16	,48	3,56	,84
1,64	18,74	2,10	1,89	2,82	2,04	,53	3,56	1,02
1,40	14,79	2,21	2,07	2,79	2,15	,52	3,49	1,04
1,78	18,86	2,00	2,08	3,14	2,60	,50	3,30	,80
1,93	15,62	2,26	2,21	2,81	2,18	,44	4,16	,92
1,53	18,56	2,14	2,00	3,16	2,22	,55	3,73	1,07

By using the procedure described in paragraph 2, the λ n likelihoods were calculated for the 448 models and are given in TABLE 3.2.

TABLE 3.2
LIKELIHOODS FOR MODEL SELECTION : $\lambda n L$

INDEPENDENT VARIABLES	DEPENDENT VARIABLES						
	123	12	13	23	1	2	3
123456	-43,6454	-32,4951	5,93117	-55,9419	14,9427	-45,8012	-8,24199
12345	-45,0927	-31,6582	3,97895	-56,9910	15,1829	-45,0715	-9,92148
12346	-41,5001	-31,3450	6,84604	-53,9126	14,8818	-44,4808	-7,59459
12356	-42,0446	-32,4046	7,75229	-55,2954	15,2858	-46,5186	-6,79962
12456	-42,3122	-32,5957	6,40192	-54,2776	13,9846	-45,1165	-7,25756
13456	-50,9745	-39,6905	8,29138	-62,1232	8,45643	-49,3577	-11,9189
23456	-43,5315	-32,8276	7,49767	-57,5265	16,1802	-47,7634	-8,03265
1234	-42,9286	-30,4553	4,94857	-55,0436	15,2165	-43,8422	-9,28312
1235	-43,4430	-31,7225	5,56006	-55,7796	15,1446	-45,4353	-8,37589
1236	-40,0787	-31,2384	8,45481	-53,3442	15,1717	-44,9760	-6,35994
1245	-47,9512	-35,9213	8,27213	-56,5710	10,8562	-45,8617	-8,89165
1246	-37,6514	-29,3666	9,23138	-51,5380	15,6326	-43,5801	-6,25501
1256	-40,9196	-32,6739	8,24280	-53,5855	14,3701	-45,7409	-5,83852
2345	-50,1847	-35,3744	1,40871	-63,6484	16,5188	-50,4855	-13,5746
2346	-43,8943	-33,1628	5,86242	-67,9541	14,7942	-47,0031	-9,81441
2356	-42,2683	-32,9538	9,04978	-57,0965	16,4601	-48,6575	-6,81200
3456	-52,3081	-41,6892	-0,510810	-64,7727	8,59413	-52,6736	-12,7945
1345	-52,7926	-39,7821	-3,48608	-60,6988	7,84963	-48,0890	-11,8509
1346	-48,0288	-38,1817	2,02789	-60,1074	9,89046	-48,6665	-10,4444
1456	-49,1725	-39,2267	1,58126	-60,9092	9,42243	-49,1726	-10,5397
1356	-48,5392	-38,5695	1,20950	-60,8969	9,16768	-49,4920	-10,5301
2456	-45,7535	-35,4942	4,24885	-58,1645	12,5018	-47,2441	-9,28204
123	-41,3236	-30,3921	6,45936	-53,8782	15,2139	-44,0108	-7,89303
124	-47,4002	-33,8066	-0,613105	-57,9553	12,0743	-45,4433	-11,9142
125	-46,7559	-36,3744	2,01792	-54,7235	10,4479	-45,4939	-7,40152
126	-36,4142	-29,3372	10,9225	-51,0005	16,0350	-44,0771	-5,03496
134	-49,6424	-37,9221	-0,487323	-59,0476	9,50778	-47,4918	-10,6504
135	-50,5408	-38,8170	-1,72594	-59,6506	8,21967	-48,3147	-10,4006
136	-45,7385	-37,0229	3,78696	-58,9942	10,5068	-48,6782	-9,32888
145	-54,9283	-42,5242	-4,25173	-60,4738	7,46711	-48,9288	-11,2471
146	-44,5836	-36,5232	4,67405	-58,0350	10,9592	-48,0722	-9,12387

TABLE 3.2 (cont.)

INDEPENDENT VARIABLES	DEPENDENT VARIABLES						
	123	12	13	23	1	2	3
156	-46,8285	-38,1596	3,55681	-59,6083	10,1356	-49,2197	-9,20027
234	-56,8120	-39,0616	-6,34779	-70,6945	15,9167	-53,9997	-21,9456
235	-48,3458	-34,8347	2,98474	-62,0815	16,5697	-50,1400	-12,0404
236	-43,2476	-33,7345	6,93104	-58,1115	14,7489	-48,2331	-8,97483
245	-52,8024	-39,2446	-2,33887	-61,6759	11,5046	-49,3545	-12,5105
246	-42,6216	-33,4579	5,57201	-56,6771	13,1437	-46,1140	-9,58912
256	-45,1857	-36,2738	5,18313	-58,2507	12,2438	-48,6175	-8,40856
345	-58,4469	-44,6221	-6,66648	-66,7081	9,18038	-53,8813	-16,0104
346	-49,6432	-40,0795	0,678147	-63,4971	8,89401	-51,7823	-13,1355
356	-50,3439	-40,9333	1,10615	-63,8988	9,07347	-53,0685	-11,7626
456	-51,4165	-41,8669	-0,596267	-63,9718	7,29487	-51,7848	-12,9379
12	-46,3909	-34,2773	0,487313	-56,1482	11,4735	-44,7876	-10,4413
13	-47,4134	-36,9171	1,14626	-57,9736	9,80482	-47,5637	-9,35568
14	-54,4201	-41,3337	-5,72228	-60,0052	7,63494	-48,0043	-12,3951
15	-52,8867	-41,5883	-2,87942	-58,8930	7,55691	-48,5457	-9,85247
16	-42,4831	-35,4990	6,43468	-56,9332	11,6029	-48,0786	-8,02388
23	-55,3574	-38,5401	-5,01745	-69,6060	15,8179	-53,8651	-20,6632
24	-59,0935	-41,9423	-10,4881	-69,6972	13,2574	-53,8972	-22,1079
25	-52,0501	-39,7294	-1,71239	-60,2964	10,6855	-49,3111	-11,0987
26	-42,4462	-34,4371	6,22782	-57,1657	12,7463	-47,6522	-9,04742
34	-61,1503	-44,5611	-11,0902	-71,3600	9,98751	-55,7882	-22,2289
35	-56,1904	-43,2859	-5,04355	-65,4155	9,48245	-53,5597	-14,6932
36	-48,1618	-39,6762	1,91687	-63,1211	9,10680	-52,5209	-12,4871
45	-58,4149	-45,6455	-7,36623	-64,9765	7,25791	-52,7386	-15,1124
46	-47,4745	-39,4820	1,57474	-61,7290	8,27493	-50,7355	-12,4791
56	-49,8882	-41,5778	0,575352	-63,4040	7,36007	-52,5546	-12,1914
1	-52,7353	-40,8037	-4,55393	-58,4641	7,37479	-47,5170	-10,9395
2	-58,4647	-42,2299	-9,82103	-68,2761	12,4760	-53,4214	-20,6877
3	-59,1877	-43,6228	-9,51965	-70,3809	10,1448	-55,7516	-21,0912
4	-63,4167	-47,3554	-14,5778	-70,0462	8,15263	-55,1362	-22,1259
5	-56,8227	-45,0432	-6,46593	-63,7621	6,85816	-52,6805	-13,9047
6	-46,3836	-39,4657	2,44678	-61,5839	8,15413	-51,7205	-12,0869
0	-62,0594	-47,0374	-13,6213	-68,8143	7,68108	-54,8140	-20,8056

proper model comparisons can only be made for models within columns in Table 3.2 and not over rows or over the whole table. Since $\ln L(\Omega_{123:126}) = -36,4142$ and $\ln L(\Omega_{123:1246}) = -37,6514$ are the two largest values in the first column it is clear that if one wants to predict Y_1, Y_2 and Y_3 , then models Z_1, Z_2, Z_6 and Z_1, Z_2, Z_4, Z_6 will be the best, but if one only knew Y_1 then model Z_2, Z_3, Z_5 seems best. The best models for the other dependent variables are underlined in Table 3.2.

As mentioned before, model selection using the method of predictive sample reuse can also be extended to the case of predicting a subset of future Y values using all the available Y data. For this case equation (2.4) will be a marginal predictive density; this type of problem, however, was not considered for the tobacco data.

4. THE ADEQUACY OF VARIABLE SUBSETS IN MULTIVARIATE REGRESSION McKAY (1979)

Denote the sample covariance matrices for the dependent variables $Y_1, \dots, Y_q$ and the independent variables $Z_1, \dots, Z_p$ by S_{yy}, S_{zz} respectively. The matrix of sample covariances between the dependent and independent variables is denoted by S_{yz} . The sample covariance matrix for the subset Ω of independent variables is a submatrix of S_{zz} denoted by $S_{\Omega\Omega}$ and the corresponding submatrix of S_{yz} is denoted by $S_{y\Omega}$.

There are many statistics appropriate for testing the adequacy of a particular subset Ω , such as the likelihood ratio statistic or the largest characteristic root statistic. The likelihood ratio test can be expressed in terms of the statistic:

$$(4.1) \quad W_{\Omega} = (1 - U_{\Omega})/U_{\Omega}$$

where

$$(4.2) \quad U_{\Omega} = |s_{yy} - s_{yz} s_{zz}^{-1} s_{yz}'| / |s_{yy} - s_{y\Omega} s_{\Omega\Omega}^{-1} s_{y\Omega}'|.$$

The hypothesis that Ω is an adequate subset is not rejected if

$$(4.3) \quad W_{\Omega} < W_{q,p-m,N-p-1}^{\alpha}$$

where m = the size of the subset of independent variables and $W_{q,p-m,N-p-1}^{\alpha}$ is the upper 100α per cent point of the distribution of $W_{q,p-m,N-p-1}$.

When all subsets Ω are to be examined for adequacy, steps should be taken to protect the Type 1 probability error rate. This can be done in the following way (Spjøtvoll (1977) and McKay (1979)):

Let A_{α} be the set of all subsets Ω satisfying the condition $W_{\Omega} < W_{q,p-m,N-p-1}^{\alpha}$ and let B_{α} be the set of all subsets Ω satisfying the condition:

$$(4.4) \quad W_{\Omega} < W_{q,p,N-p-1}^{\alpha} = W_{q,p-m,N-p-1}^{\alpha_m}$$

Let the set A_{α}^{+} consists of all the subsets in A_{α} and extensions of these which are obtained by adding more independent variables. It is easy to see that $A_{\alpha} \subseteq A_{\alpha}^{+} \subseteq B_{\alpha}$ and that A_{α}^{+} contains all adequate subsets with probability no less than $1-\alpha$. (Note, however, that the probability that the set A_{α} contains all adequate subsets is generally less than $1-\alpha$. B_{α} on the other hand will contain all adequate subsets with probability at least $1-\alpha$ but may be unrealistically large.)

Complex situations can often be clarified by means of appropriate graphical displays. On a P_Ω plot, where

$$(4.5) \quad P_\Omega = P(W_\Omega \geq \text{the observed value of } W_\Omega),$$

A_α consists of all subsets above the line

$P_\Omega = \alpha$, while B_α consists of all the subsets above the curve $P_\Omega = \alpha_m$.

P_Ω can be calculated by making use of equations (11), (14), (25) and (26) on pages 195, 196 and 199 respectively of Anderson (1958) and the fact that $\{b - \frac{1}{2}(t - a + 1)\} \ln(1 + W_{t,a,b}) \sim \tilde{C}x_{ta}^2$. x_{ta}^2 is a chi-squared distribution with ta degrees of freedom and $\tilde{C}$ is a correction factor as given by Schatzoff (1966).

By applying a simultaneous test procedure of level $\alpha = 0.05$ to the tobacco data, a subset Ω belongs to the set $B_{0,05}$ if

$$(4.6) \quad W_\Omega < W_{3,6,18}^{0,05} = 3,658$$

$B_{0,05}$ contains all adequate subsets with probability at least 0,95. A subset Ω belongs to $A_{0,05}$ if

$$(4.7) \quad W_\Omega < W_{3,6-m,18}^{0,05}; \text{ i.e. if } P_\Omega > 0,05; \text{ for } m = 1, 2, 3, 4 \text{ and } 5. \text{ It is clear that all members of } A_{0,05} \text{ are also members of } B_{0,05}.$$

Table 4.1 gives the statistics for the determination of the adequate subsets for the tobacco data. An asterisk indicates a member of $A_{0,05}$, while underlined subsets indicate members of $B_{0,05}$. For lack of space these statistics are only calculated for the whole set of dependent variables and not for subsets of Y_1 , Y_2 and Y_3 .

TABLE 4.1

STATISTICS FOR THE DETERMINATION OF ADEQUATE SUBSETS

Ω	m	W_{Ω}	P_{Ω}	Ω	m	W_{Ω}	P_{Ω}
12	2	3,8275	0,0053178	245	3	4,3529	0,0006071
13	2	4,0044	0,0042814	246	3	1,6962	0,044539
14	2	8,3429	0,000077	256	3	1,8765	0,0307847
15	2	6,5411	0,0003204	345	3	7,7521	0,0000201
16	2	2,5252	0,0320323	346	3	3,6581	0,0015317
23	2	8,4124	0,0000732	356	3	3,5736	0,0017272
24	2	12,4917	0,0000062	456	3	4,0582	0,0008873
25	2	5,7263	0,0006737	1234*	4	0,8353	0,11274
26	2	2,4624	0,0352772	1235*	4	0,8741	0,09982
34	2	14,6362	0,0000022	1236*	4	0,4343	0,40994
35	2	8,9152	0,0000515	1245	4	1,6258	0,011975
36	2	4,4152	0,0026439	1246*	4	0,3213	0,57923
45	2	10,6003	0,0000176	1256*	4	0,4992	0,33308
46	2	4,4348	0,0025856	2345	4	2,2361	0,002904
56	2	4,9009	0,0015491	2346*	4	1,0940	0,05119
123*	3	1,2308	0,122876	2356*	4	0,7211	0,16225
124	3	2,8439	0,0052947	3456	4	3,0260	0,0006266
125	3	2,2719	0,0143582	1345	4	3,0156	0,00063814
126*	3	0,6029	0,5110771	1346	4	1,8448	0,007006
134	3	3,4306	0,0021261	1456	4	2,1026	0,0038816
135	3	3,6027	0,001657	1356	4	1,8847	0,0063767
136	3	2,1442	0,0182411	2456*	4	1,3041	0,280636
145	3	5,5187	0,0001583	12345*	5	0,5517	0,06477
146	3	2,1934	0,016622	12346*	5	0,1824	0,4299
156	3	2,4393	0,0105924	12356*	5	0,2039	0,3828
234	3	6,8093	0,0000449	12456*	5	0,2441	0,3082
235	3	2,8259	0,0054541	13456	5	1,5682	0,00143
236*	3	1,6408	0,0500304	23456*	5	0,4277	0,1183

A visual impression of the values given in Table 4.1 is obtained by plotting the significance probability P_{Ω} against m in Figure 4.1. The values for α_m are given in Table 4.2. These values are used to draw the broken curve in Figure 4.1

TABLE 4.2

VALUES OF α_m FOR $\alpha = 0,05$

m	2	3	4	5
m	0,0065817	0,0015319	0,000219986	0,00001359

It is clear that α_m is very small when m is large.

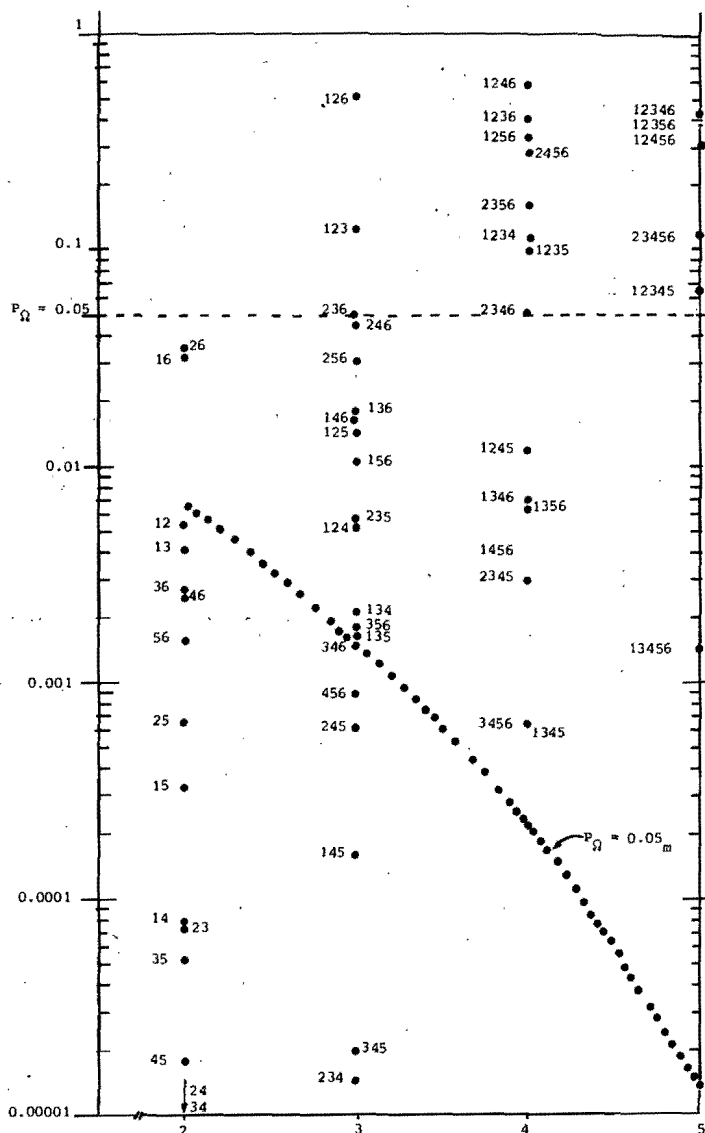


Figure 4.1 P_{Ω} plot for examining the adequacy of subsets of Tobacco Data.
Note: The Vertical scale is logarithmic.

It follows that $A_{0,05}$ consists of the subsets {123}, {126}, {236}, {1234}, {1235}, {1236}, {1246}, {1256}, {2346}, {2356}, {2456}, {12345}, {12346}, {12356}, {12456}, {23456}. These are denoted by points above the line

$P_{\Omega} = 0,05$ in Figure 4.1. Points above the broken curve

$P_{\Omega} = 0,05_m$ represent members of $B_{0,05}$

Since $A_{0,05}^+$ and $A_{0,05}$ are identical for this problem, the adequate subsets are those given above.

The two largest probabilities 0,511077 and 0,57923 give an indication that the two best subsets may be {126} and {1246}. Note that these two subsets were also selected as best by the method of predictive sample reuse.

5. MALLOWS' C_{p^*} STATISTIC (1973) AND ALLEN'S PRESS (1971)

In the case of multivariate regression analysis Mallows' C_{p^*} statistic can be defined as:

$$(5.1) \quad C_{p^*} = \frac{1}{q} \text{tr} \{S^{-1} \cdot \text{RSS}_{p^*} - (N-2p^*)I_q\}$$

where S is an estimate of the residual covariance matrix and RSS_{p^*} is the residual sums of squares and crossproducts matrix from a model containing p^* parameters in the model including α_0 . Table 5.1 gives the various C_{p^*} values for the tobacco data. To save space only the "best" subsets for each of $p^* = 1, 2, 3, 4, 5, 6$ and 7 are given.

TABLE 5.1

VALUES OF C_{p^*} AND p^* FOR THE TOBACCO DATA

p^*	1	2		3		4		5		6		7
Ω	-	6	1	26	12	126	123	1246	1236	12346	12356	123456
C_{p^*}	50,78	22,17	23,16	10,72	15,37	4,15	6,73	4,86	5,39	6,09	6,22	7

From Table 5.1 it is clear that the C_{p^*} statistic selects subsets 126 and 1246 as the two "best" models for predicting Y_1 , Y_2 , Y_3 . (For adequate models $C_{p^*} \approx p^*$.)

The basic calculations for the PRESS (predictive discrepancy sum of squares) procedure are:

Delete the first set of observations on the dependent and independent variables.

Fit all possible regression models to the remaining $N-1$ data points.

Use each fitted model to predict $\underline{Y}_1$ by $\hat{\underline{Y}}_{1p^*}$ (say).

Repeat the previous steps to predict $\underline{Y}_2$ by $\hat{\underline{Y}}_{2p^*}$, and so on till $\underline{Y}_N$ by $\hat{\underline{Y}}_{Np^*}$.

For each subset regression model calculate:

$$(5.2) \text{ PRESS} = \sum_{i=1}^N (\underline{Y}_i - \hat{\underline{Y}}_{ip^*})' S^{-1} (\underline{Y}_i - \hat{\underline{Y}}_{ip^*}).$$

The "best" subset regression model will have a comparatively small PRESS but not involve too many independent variables. Table 5.2 gives the PRESS values for the tobacco data. Only the "best" subsets for each p^* are given.

TABLE 5.2
VALUES OF PRESS FOR THE TOBACCO DATA

p*	1	2		3		4		5		6		7
Ω	-	6	1	26	16	126	123	1246	1236	12346	12456	123456
PRESS	240,17	151,38	157,14	115,23	115,55	89,62	100,12	90,91	96,25	97,42	103,71	105,04

From Table 5.2 it follows that Allen's PRESS procedure selects subsets 126 and 1246 as the best models.

6. CONCLUSION

From our results it is clear that the two "best" models for the set of dependent variables $\{Y_1, Y_2, Y_3\}$ are 126 and 1246: They were selected as best by all four methods: Allen's PRESS, Mallows' C_{p^*} statistic and the "Predictive Sample Reuse" method selected 126 as best and 1246 as second best while McKay's procedure selected them in reverse order.

The similarity between the method of predictive sample reuse and Mallows' C_{p^*} statistic is not surprising because it was mentioned by Geisser and Eddy (1980) in a letter to JASA that these two methods are asymptotically equivalent for the multiple normal regression case in ordering subsets.

ACKNOWLEDGEMENT

The authors wish to thank the referees for their helpful comments in revising this paper.

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